Method: from GDP to oscillators

How each economy's GDP becomes a phase oscillator and how the Kuramoto order parameter $r(t)$ measures synchronization, with the first evidence that the measured synchronization is not chance. The four companion-book figures use Maddison real GDP per capita, the paper's data; one of them (F1.4) draws a series that matches no stored results file and should be redrawn. The six first-pass figures used World Bank GDP in current dollars, which mixes real output with inflation and exchange rates; they are kept as the record of the first run.

Status key

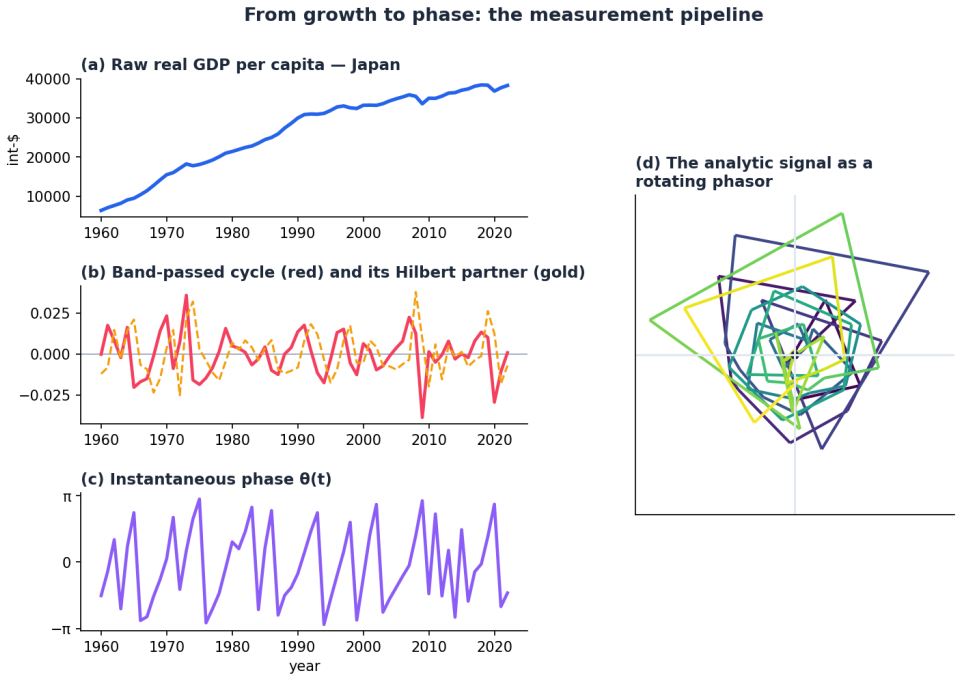
CURRENT	Paper data (Maddison 2023) and the paper's conventions; numbers match docs/methods.md.
NEEDS RELABELLING	Numbers right, but an annotation on the figure is out of date (said in the caption).
CONVENTION W	Hilbert transform applied to 1970–2022 only (convention W); the paper uses P.
SUPERSEDED	Earlier data or crisis definition; the record of how the paper got here.
DESCRIPTIVE	Atlas material from the Global Macro Database; describes, does not test.
UNVERIFIED SERIES	The series drawn matches no stored results file; redraw before use.

In this book

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F1.1 CURRENT

From growth to phase: the measurement pipeline (Japan)

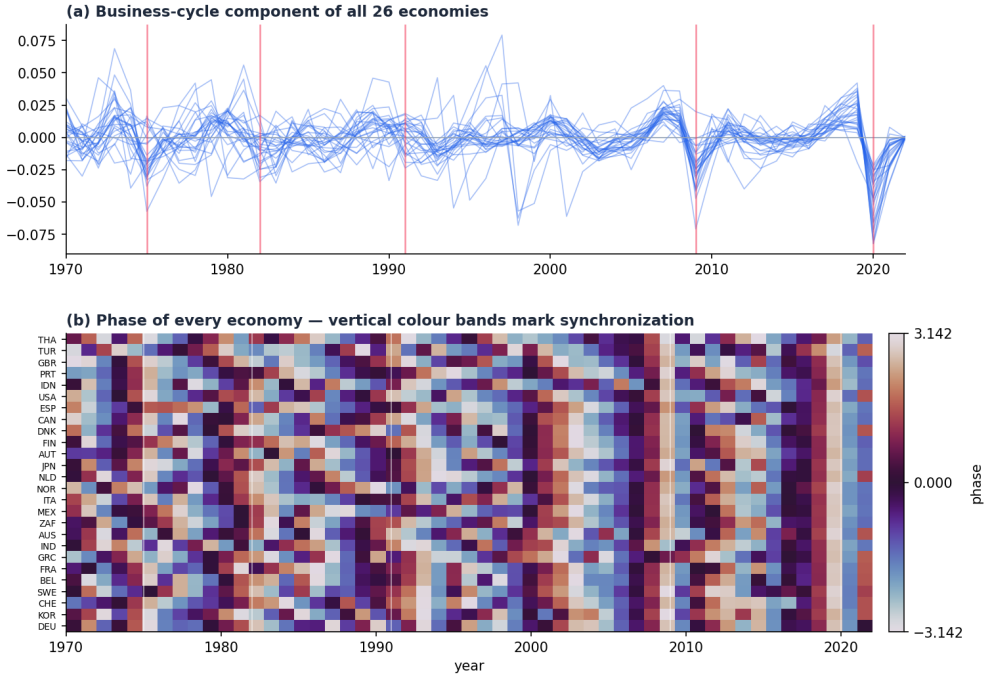


(a) Japan's real GDP per capita rises smoothly; on its own it is not an oscillator. (b) The 2–9 year Butterworth band-pass (order 2, forward–backward) leaves the business-cycle wobble (red); its Hilbert partner (gold) is the same wobble shifted by a quarter cycle. (c) The instantaneous phase $\theta(t) = \arg z(t)$ sweeps from $-\pi$ to π once per cycle. (d) The analytic signal $z(t)$ drawn in the complex plane: a rotating phasor whose angle is the phase and whose length is the amplitude $A(t)$.

Numbers Illustration of Methods §2; no test.
Made by Companion-book figure A1 (plotting script lost with the September workspace) · **Data** Maddison 2023, Japan, 1960–2022
File figures/01_method/F1-01_pipeline_japan_real.png

F1.2 CURRENT

All 26 economies as oscillators

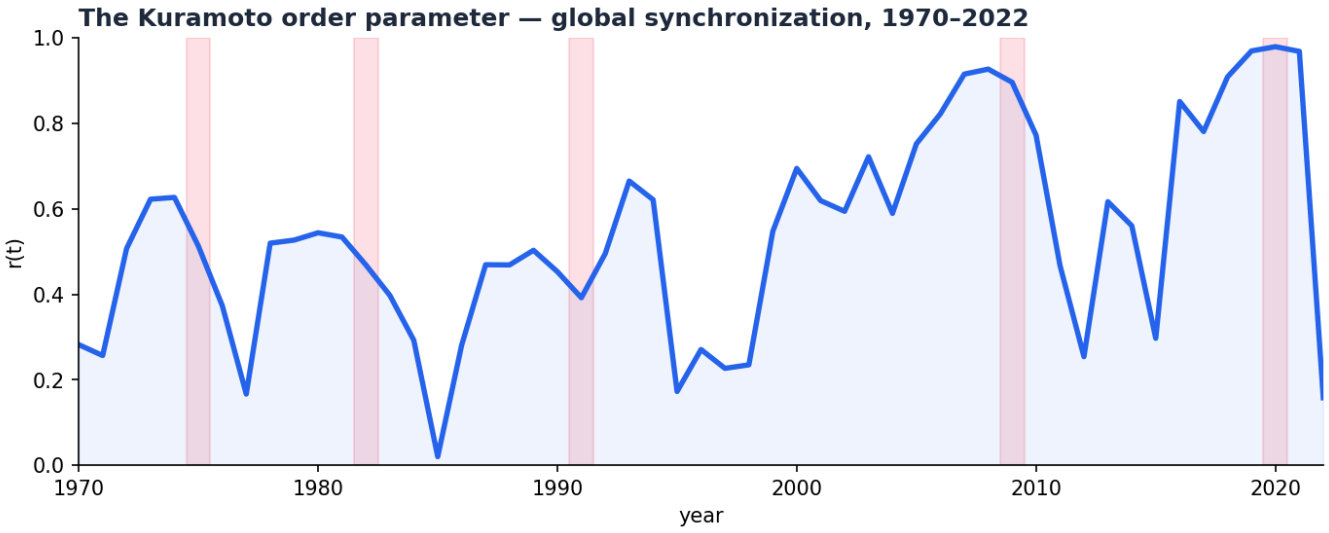


(a) The band-passed cycle of every economy, with the five global recession years in red. (b) The phase of every economy by year, one row per economy, colour giving the position in the cycle. A vertical stripe of one colour is a year in which the economies share a phase; the clearest stripes are around 2009 and 2020.

Numbers Descriptive.
Made by Companion-book figure A2 · **Data** Maddison 2023, 26 economies, 1970–2022
File figures/01_method/F1-02_all_26_oscillators.png

F1.3 CURRENT

The Kuramoto order parameter $r(t)$, 1970–2022

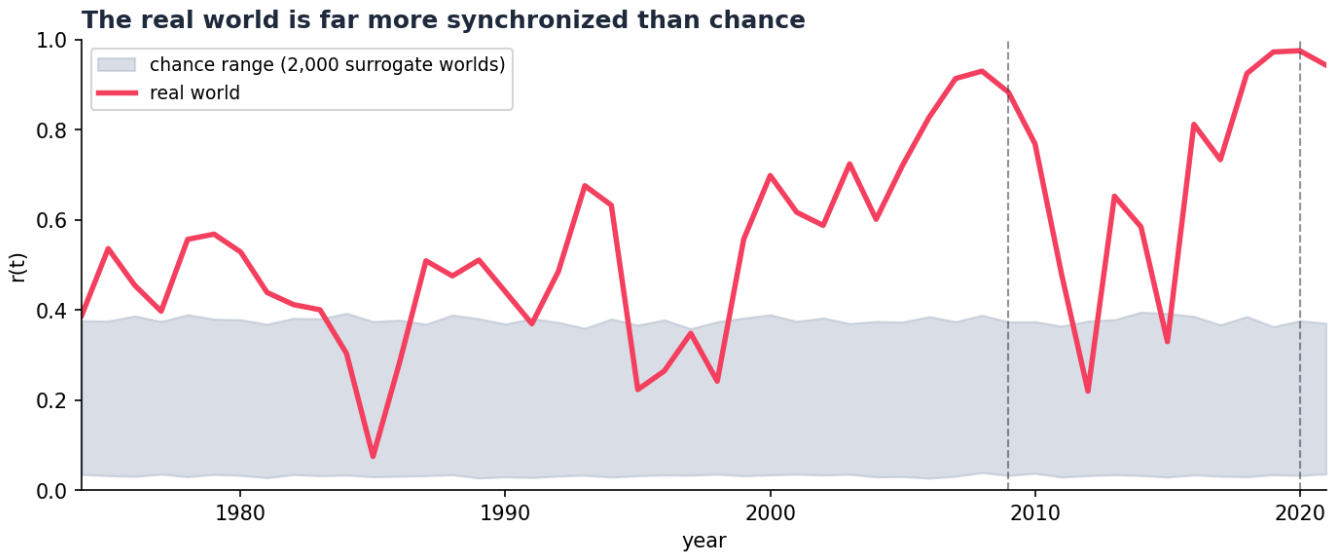


$r(t) = \left|\frac{1}{N}\sum_i e^{\mathrm{i}\theta_i(t)}\right|$ for the 26 economies, computed with the paper's convention P (filter and Hilbert transform on 1950–2022, statistics on 1970–2022). Pink bands are the five global recessions. With 26 independent oscillators the expected r is about 0.20, so only the variation over time is interpreted.

Numbers Lowest r = 0.02 (1985); highest r = 0.98 (2020).
Made by Companion-book figure A3; the series is r_by_year in paper1/results/kose_terrones_results.json · **Data** Maddison 2023
File figures/01_method/F1-03_order_parameter_real.png

F1.4 UNVERIFIED SERIES

The real world is far more synchronized than chance

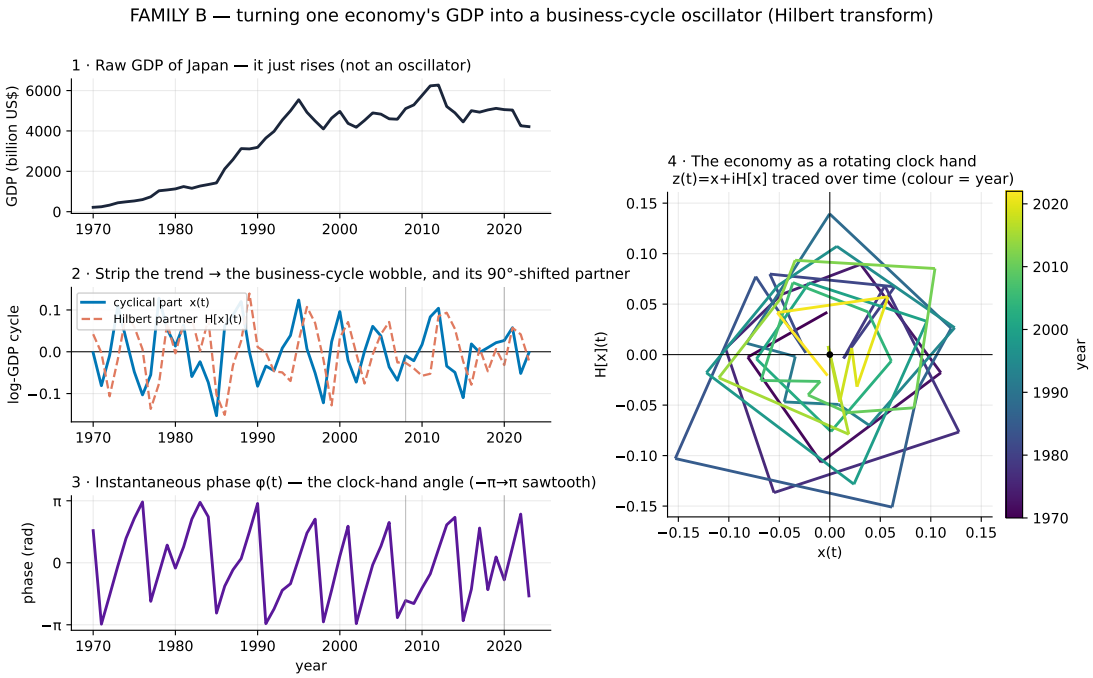


The observed $r(t)$ (red) against the range produced by surrogate worlds in which every economy keeps its own power spectrum but its phases are randomized independently, so all alignment is destroyed (grey). This companion rendering used 2,000 surrogates; the paper's surrogate tests use 5,000. Its red line matches neither stored $r(t)$ series (it shows about 0.40 in 1977, where convention P gives 0.17 and W 0.14), so the series behind it cannot be identified. Redraw from the convention-P series with 5,000 surrogates before using it.

Numbers As drawn: observed r above the chance range in most years and far above it in 2009 and 2020 (not verified).
Made by Companion-book figure A4 · **Data** Maddison 2023
File figures/01_method/F1-04_real_vs_chance_real.png

F1.5 SUPERSEDED

First pass: the Hilbert pipeline on current-dollar GDP (Japan)

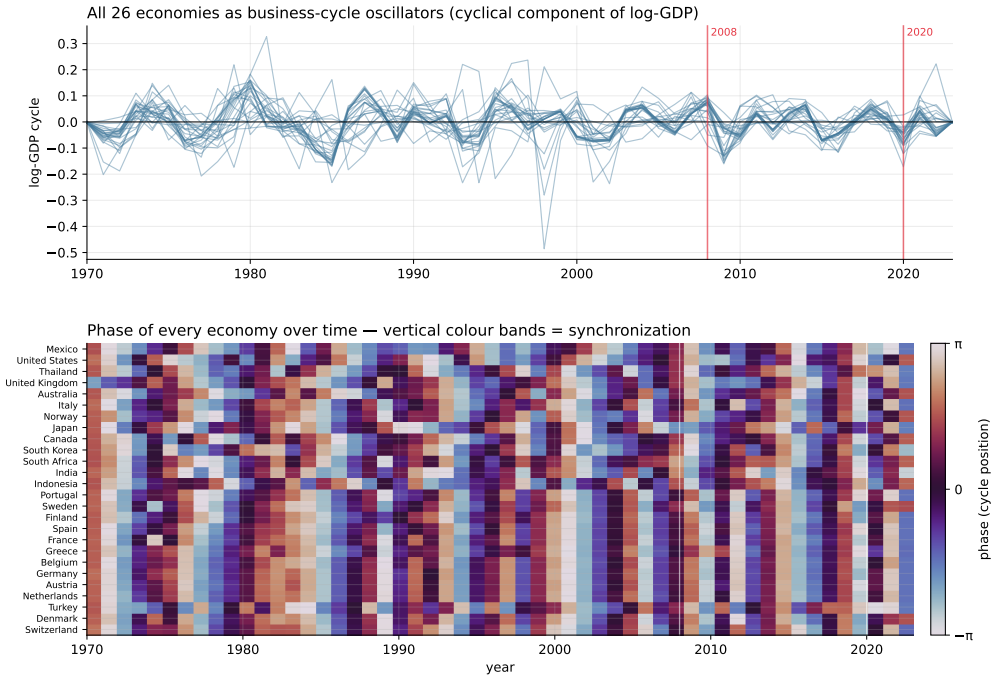


The same four steps as F1.1, run in July on World Bank GDP in current US dollars (1970–2023). Nominal dollar GDP mixes real output, inflation and exchange-rate swings, which is why the paper moved to Maddison real GDP per capita.

Numbers Illustration only.
Made by paper1/scripts/eraA_claude_ai/hilbert_oscillators.py (reproduces hilbert_output.csv exactly) · **Data** World Bank NY.GDP.MKTP.CD via datasets/gdp
File figures/01_method/F1-05_pipeline_japan_current_usd.pdf

F1.6 SUPERSEDED

First pass: all 26 economies on current-dollar GDP

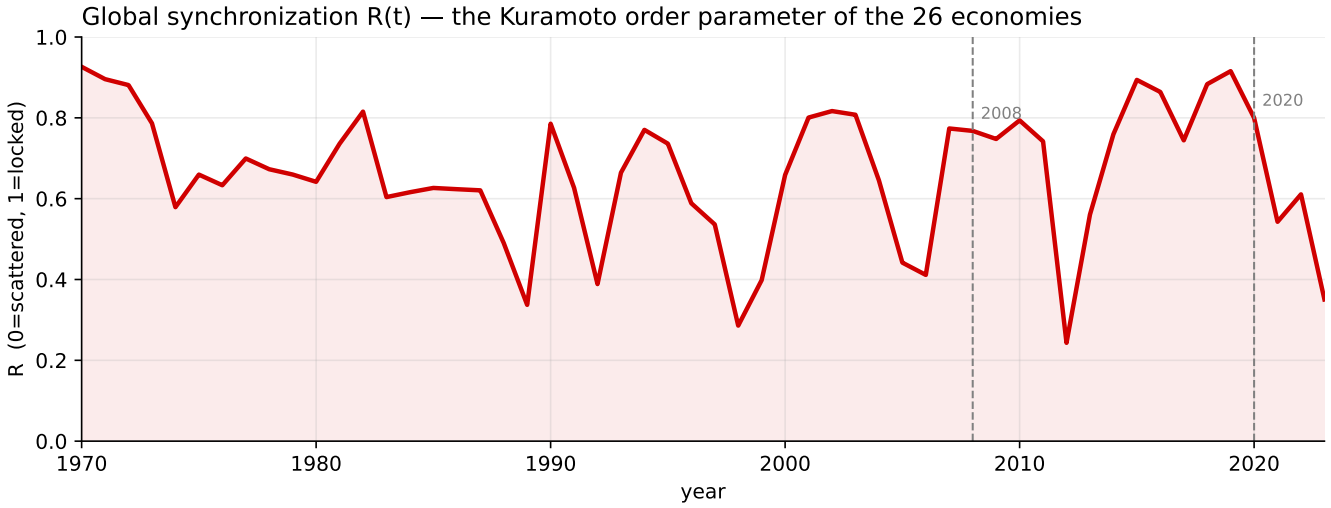


Every economy's cyclical component and the phase heat map, from the current-dollar first pass. Red lines mark 2008 and 2020.

Numbers Descriptive.
Made by hilbert_oscillators.py · **Data** World Bank current US$
File figures/01_method/F1-06_all_26_current_usd.pdf

F1.7 SUPERSEDED

First pass: $R(t)$ on current-dollar GDP, 1970–2023

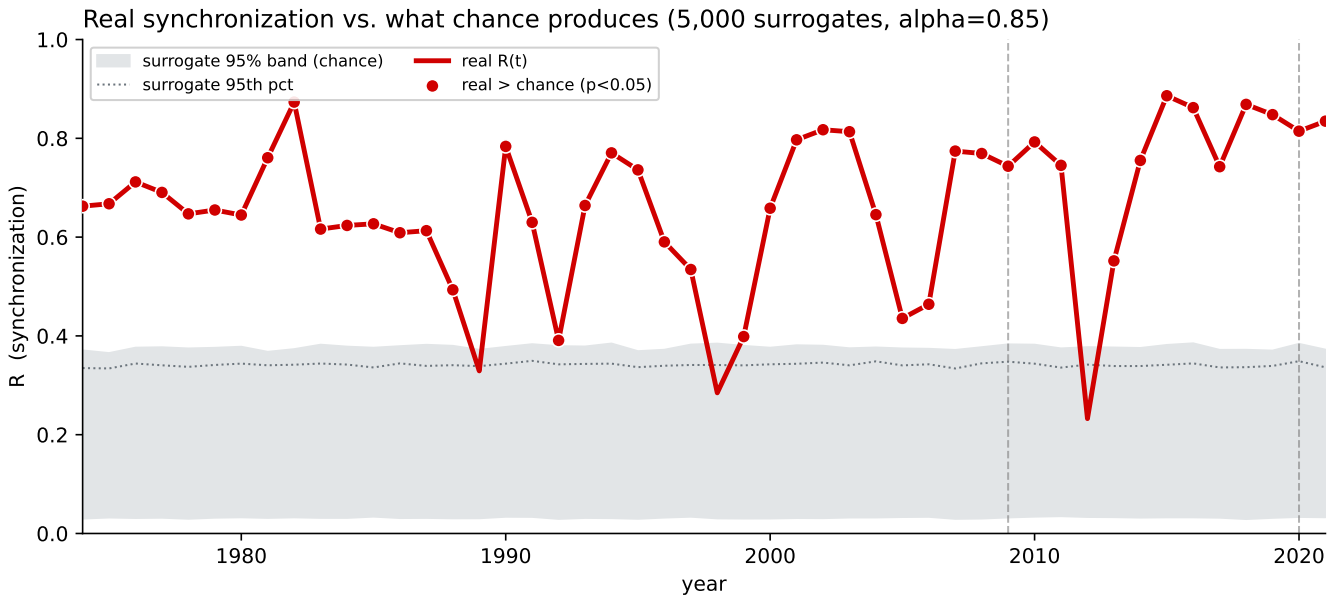


The order parameter from the first pass. Its level differs a lot from the real-GDP series (for example 0.72 against 0.28 in 1970) because exchange-rate and inflation swings move nominal dollar GDP together across countries.

Numbers Peak 0.91 (2019); trough 0.24 (2012), from crisis_results.json.
Made by hilbert_oscillators.py · **Data** World Bank current US$
File figures/01_method/F1-07_order_parameter_current_usd.pdf

F1.8 SUPERSEDED

First pass: real synchronization against 5,000 surrogates

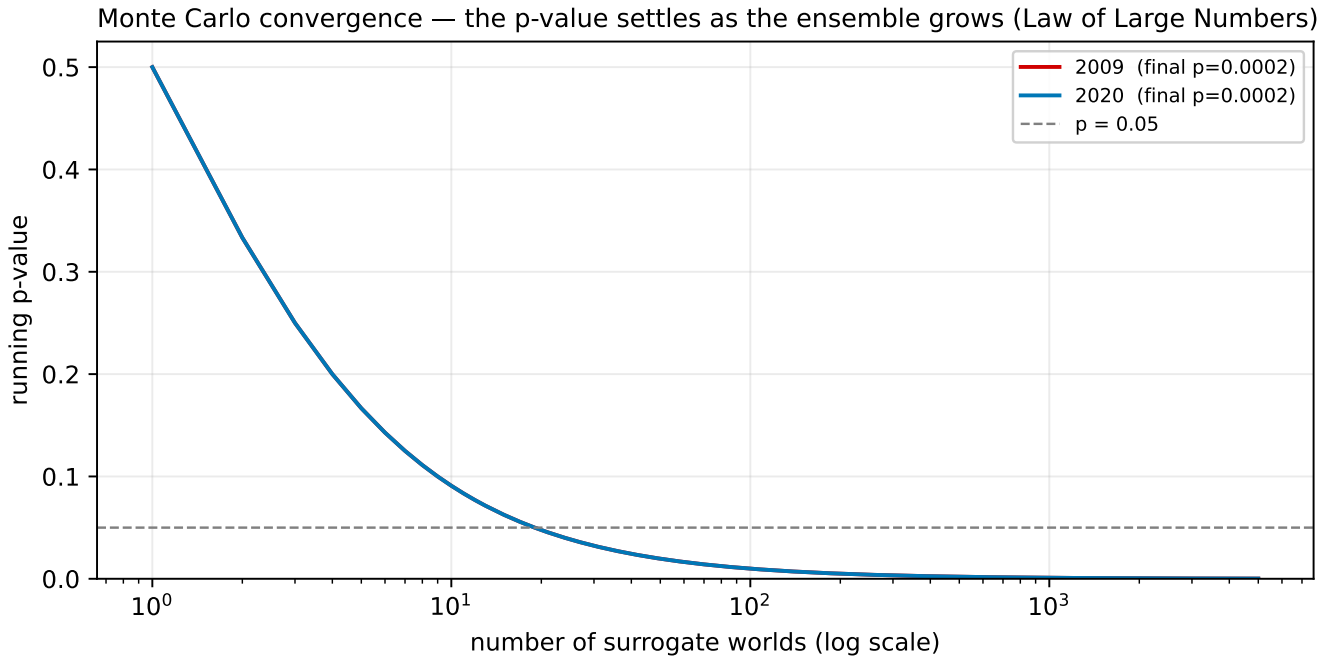


Observed $R(t)$ against the 95% band of 5,000 spectrum-preserving surrogate economy sets; dots mark years above the surrogate 95th percentile. The method is the paper's; the data are the superseded current-dollar series.

Numbers 2009 and 2020 at the surrogate floor, p = 0.0002.
Made by paper1/scripts/eraA_claude_ai/surrogate_test.py (reproduces surrogate_results.csv exactly) · **Data** World Bank current US$
File figures/01_method/F1-08_surrogate_band_current_usd.pdf

F1.9 SUPERSEDED

First pass: Monte Carlo convergence of the surrogate p-value

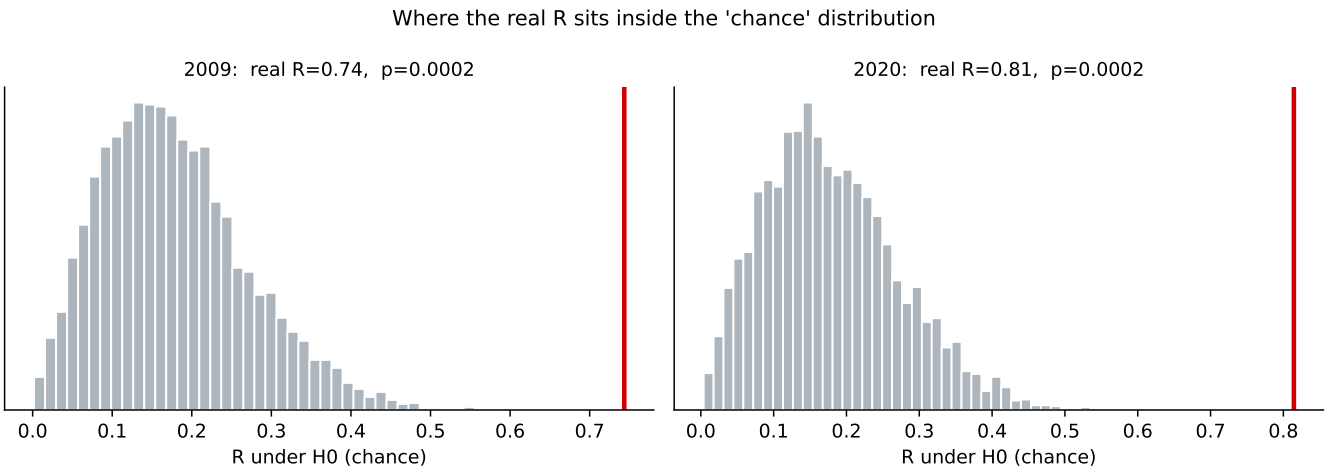


The running p-value for 2009 and 2020 as the number of surrogate worlds grows to 5,000. It settles at the floor 1/5001, showing that 5,000 draws are enough for a p-value of this size.

Numbers Final p = 0.0002 for both years.
Made by surrogate_test.py · **Data** World Bank current US$
File figures/01_method/F1-09_surrogate_convergence.pdf

F1.10 SUPERSEDED

First pass: where the real R sits in the chance distribution



Histograms of R under the no-coupling null for 2009 and 2020, with the observed value in red, far in the tail.

Numbers 2009: R = 0.74, p = 0.0002. 2020: R = 0.81, p = 0.0002.
Made by surrogate_test.py · **Data** World Bank current US$
File figures/01_method/F1-10_null_histograms.pdf